

10 Automation Scripts for Monitoring & Logging

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System Resource Monitoring (Bash + Cron)

```
#!/bin/bash
LOG_FILE="/var/log/system_monitor.log"
echo "=== $(date) ===" >> $LOG_FILE
echo "CPU Usage:" >> $LOG_FILE
top -bn1 | grep "Cpu(s)" | awk '{print $2 + $4}' >> $LOG_FILE
echo "Memory Usage:" >> $LOG_FILE
free -h >> $LOG_FILE
echo "Disk Usage:" >> $LOG_FILE
df -h >> $LOG_FILE
echo "===== " >> $LOG_FILE
```

Docker Container Health Check (Python)

```
import docker
import datetime

client = docker.from_env()
log_file = "/var/log/docker_health.log"

with open(log_file, "a") as log:
    log.write(f"=== {datetime.datetime.now()} ===\n")
    for container in client.containers.list():
        log.write(f"{container.name} - {container.status}\n")
    log.write("=====\n")
```

Kubernetes Pod Status Monitor (Kubectl + Bash)

```
#!/bin/bash
LOG_FILE="/var/log/k8s_pod_status.log"
echo "=== $(date) ===" >> $LOG_FILE
kubectl get pods --all-namespaces --field-selector=status.phase!=Running >> $LOG_FILE
echo "===== " >> $LOG_FILE
```

Log File Rotation and Cleanup (Bash)

```
#!/bin/bash
find /var/log -name "*.log" -mtime +7 -exec rm -f {} \;
echo "Log cleanup completed at $(date)" >> /var/log/cleanup.log
```

AWS EC2 Instance Health Check (AWS CLI + Bash)

```
#!/bin/bash
LOG_FILE="/var/log/aws_ec2_health.log"
echo "=== $(date) ===" >> $LOG_FILE
aws                ec2                describe-instance-status            --query
"InstanceStatuses[*].{ID:InstanceId,Status:InstanceState.Name}"    --output    table    >>
$LOG_FILE
echo "======" >> $LOG_FILE
```

Jenkins Job Failure Alert (Python)

```
import requests

JENKINS_URL = "http://your-jenkins-url/api/json"
USERNAME = "your-username"
PASSWORD = "your-password"

response = requests.get(JENKINS_URL, auth=(USERNAME, PASSWORD))
data = response.json()

for job in data["jobs"]:
    if job["color"] == "red":
        print(f"Job {job['name']} failed!")
```

Log Anomaly Detection (Python)

```
import re

log_file = "/var/log/syslog"
error_keywords = ["error", "fail", "critical"]

with open(log_file, "r") as log:
    for line in log:
        if any(word in line.lower() for word in error_keywords):
            print(f"ALERT: {line}")
```

Trivy Security Scan Automation (Bash)

```
#!/bin/bash
IMAGE="your-docker-image"
trivy image $IMAGE > /var/log/trivy_scan.log
```

ArgoCD App Health Check (Bash)

```
#!/bin/bash
LOG_FILE="/var/log/argocd_health.log"
```

```
echo "=== $(date) ===" >> $LOG_FILE
argocd app list >> $LOG_FILE
echo "===== " >> $LOG_FILE
```

Network Latency Monitoring (Bash)

```
#!/bin/bash
LOG_FILE="/var/log/network_latency.log"
PING_RESULT=$(ping -c 5 google.com | tail -1 | awk '{print $4}' | cut -d '/' -f 2)
echo "$(date): Latency = ${PING_RESULT}ms" >> $LOG_FILE
```